

Building a Simple Agent Framework 3

Let's go back to the file agent.py and build under. The original implementation uses direct function calls and a lot of conditional logic, so the agent may not make the implementation using your new framework.

Define the Goals

First, let's define a goal for our simple agent:

```
def define_goals_for_the_agent():
    """Define the agent's goals.
    Returns:
        List[str]: A list of goals for the agent.
    """
    goals = [
        "Explore the environment.",
        "Find the treasure chest.",
        "Return to the starting point.",
    ]
    return goals
```

Create Actions Using the Framework

Next, let's create our first framework properly structured (2022) in our Agent.py file:

```
def create_actions_using_framework():
    """Create actions using the framework.
    Returns:
        List[Action]: A list of actions for the agent.
    """
    actions = [
        # Action 1: Explore the environment
        Action(
            name="Explore",
            description="Explore the environment.",
            execute=lambda state: state.update({"explored": True}),
        ),
        # Action 2: Find the treasure chest
        Action(
            name="FindTreasure",
            description="Find the treasure chest.",
            execute=lambda state: state.update({"treasure_found": True}),
        ),
        # Action 3: Return to the starting point
        Action(
            name="ReturnHome",
            description="Return to the starting point.",
            execute=lambda state: state.update({"home": True}),
        ),
    ]
    return actions
```

Create and Run the Agent

Now let's put it all together:

```
def create_and_run_agent():
    """Create and run the agent.
    Returns:
        Agent: The created agent.
    """
    agent = Agent(
        name="SimpleAgent",
        goals=define_goals_for_the_agent(),
        actions=create_actions_using_framework(),
    )
    agent.run()
    return agent
```

Complete Implementation

Here's the complete implementation using the new framework:

```
def create_and_run_agent():
    """Create and run the agent.
    Returns:
        Agent: The created agent.
    """
    agent = Agent(
        name="SimpleAgent",
        goals=define_goals_for_the_agent(),
        actions=create_actions_using_framework(),
    )
    agent.run()
    return agent
```

Key Differences and Benefits

Compared to the original implementation, the new framework offers:

- 1. **Modularity:** Each component has a clear purpose and is separated from others.
- 2. **Scalability:** It can easily accommodate new actions or goals without changing the core logic.
- 3. **Flexibility:** The goals and actions can be added easily.
- 4. **Testability:** The framework allows for testing individual components in isolation.
- 5. **Maintainability:** The framework is easier to update and maintain.

This document also makes it easier to understand and maintain the code, especially as your agent grows in complexity.

Using the Agent

Here's how you can use the agent in your application:

```
def use_agent():
    """Use the agent in your application.
    Returns:
        Agent: The agent object.
    """
    agent = create_and_run_agent()
    # You can now use the agent to interact with the environment.
    # For example, you can call agent.execute(action) to perform an action.
    # Or you can call agent.get_state() to get the current state.
    return agent
```

This document provides a much more detailed, modular, and maintainable way to build your agent.